Price1-Only Ruleout: Dynamic versus Non-Dynamic Sharing

BoardPharma

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1 Comparison Design

Within each saved cohort panel, observations are grouped by company and product (`BoardName` × `product`). Using all observed quarters from the earliest through the latest quarter in that cohort file, the price1-only ruleout score for company b, product p, and cohort c is

$$R_{bpc}^{\text{price1}} = \frac{\max(\texttt{price1}_{bpc})}{\min(\texttt{price1}_{bpc})}.$$

Ruleout = 0 is the unfiltered benchmark. For threshold $\tau \in \{100, 10\}$, every row belonging to a company–product history is removed from that cohort when $R_{bpc}^{\text{price1}} > \tau$; equality is retained. The score uses original price levels and is calculated before the event-time restriction, upper-p95 winsorization, and log transformation. A cohort with no remaining treated observations is not stacked.

Pages 2–7 use the same layout. The upper half reports the three **dynamic-sharing** regressions in the order ruleout 0, 100, and 10. The lower half reports the corresponding three **non-dynamic-sharing** regressions in the same order. Treatment status, control status, event timing, and event cohort are read from the same saved cohort files and are not redefined by ruleout. Dynamic versus non-dynamic sharing changes the ATC3 sharing assignment used to separate the two plotted treatment-effect series.

All six specifications within an event-side page otherwise use the same regression design: `ever_board=1`, `ieoc_ref=1`, `late=3`, `price1`, req2, the Not control group, company–product–cohort fixed effects, cohort–calendar-quarter fixed effects, company clustering, horizons 0–7, three pre-event coefficients, and no Kappa, other-event, or ATC3 fixed-effect controls.

Table 1: Estimation observations across the six event–side regressions

Ruleout setting	Dynamic sharing	Non-dynamic sharing
0 (off)	583,584	583,584
100 (price1)	580,560	580,560
10 (price1)	551,316	551,316

Notes. Counts sum final stacked estimation observations across the six regressions, not unique company–product records.

Dissolution After a Prior Interlock

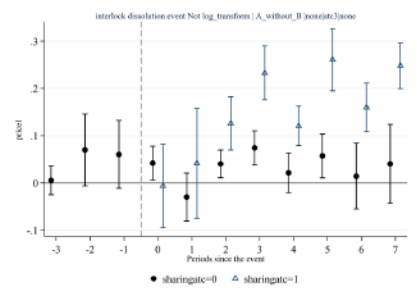
The third row on pages 2 and 3 adds `dissolution_after_interlock=1` with ruleout off. A dissolution treatment enters this sample if the same two companies had a `to_B_still_in_A` event in 2009Q1 or later and strictly before the dissolution quarter. Both events must pass the existing req2 conditions under `req1_new` and `stay_new`. Pair matching allows A and B to switch between events. The treatment side is assigned at dissolution.

The restriction is applied after the original balanced treatment and control groups are built. Retained cohorts keep their original control observations. Cohorts with no remaining treated observations are omitted. ATC3 sharing is recalculated using only qualifying counterpart companies. Dynamic sharing uses each observation quarter, while non-dynamic sharing uses the dissolution quarter throughout the window.

2 Interlock Dissolution — A-side Treatment

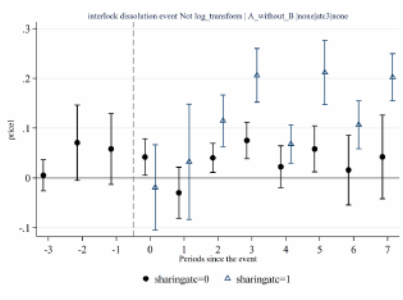
Dynamic sharing

Ruleout = 0



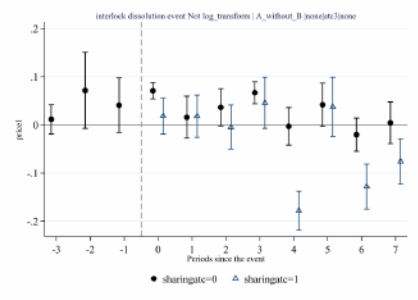
Estimation observations: 109,824

Ruleout = 100



Estimation observations: 109,236

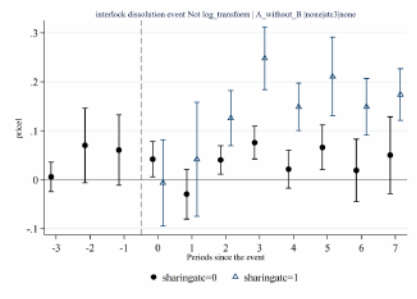
Ruleout = 10



Estimation observations: 105,708

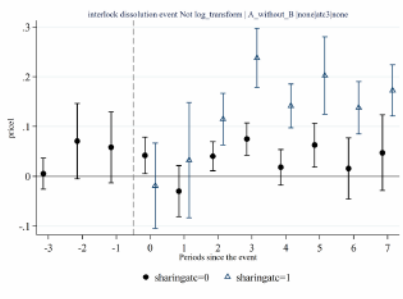
Non-dynamic sharing

Ruleout = 0



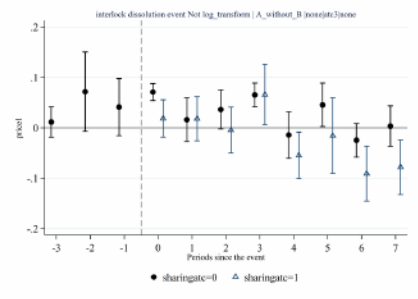
Estimation observations: 109,824

Ruleout = 100



Estimation observations: 109,236

Ruleout = 10



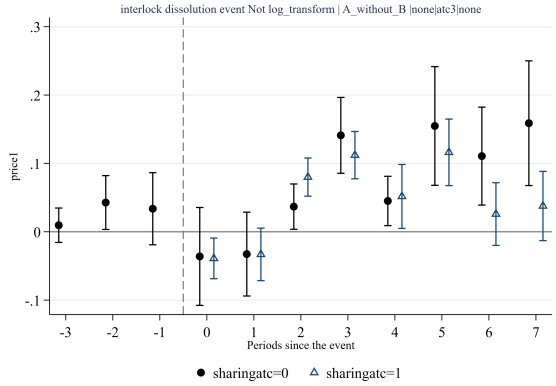
Estimation observations: 105,708

Notes. Black circles report estimates for `sharingatc=0`; hollow blue triangles report estimates for `sharingatc=1`. Error bars are 95% confidence intervals. Within each row, panels differ only in the price1 ruleout setting and resulting sample.

Dissolution after a prior interlock

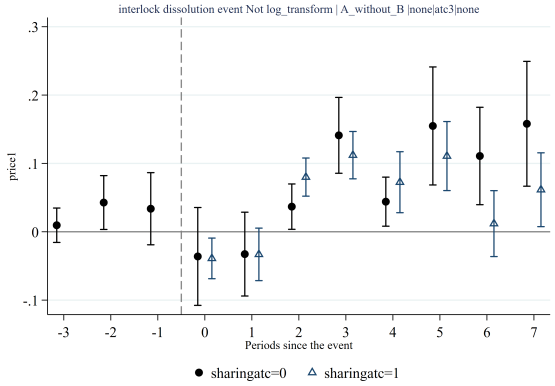
Ruleout = 0

Dynamic sharing



Estimation observations: 85,248

Non-dynamic sharing

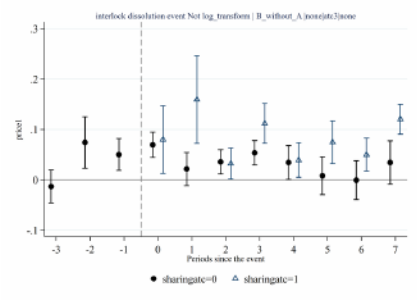


Estimation observations: 85,248

3 Interlock Dissolution — B-side Treatment

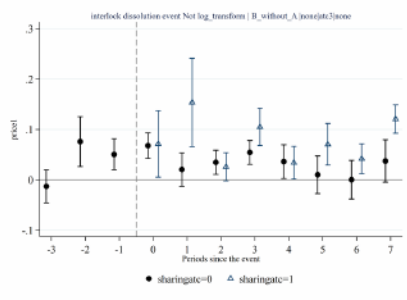
Dynamic sharing

Ruleout = 0



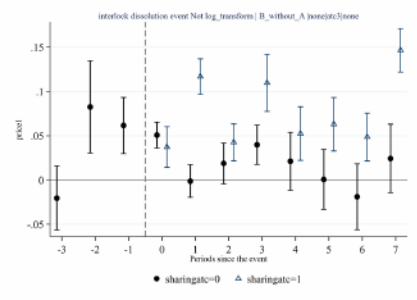
Estimation observations: 94,944

Ruleout = 100



Estimation observations: 94,440

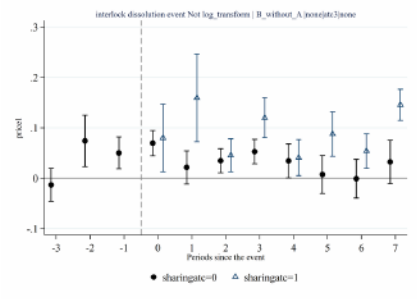
Ruleout = 10



Estimation observations: 88,332

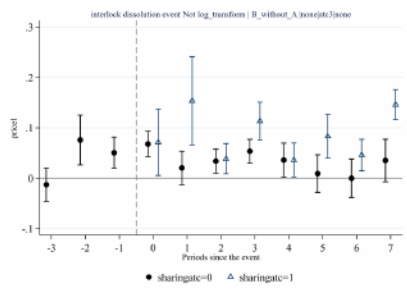
Non-dynamic sharing

Ruleout = 0



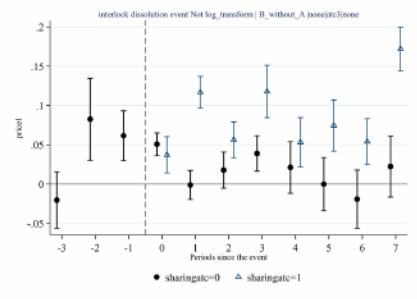
Estimation observations: 94,944

Ruleout = 100



Estimation observations: 94,440

Ruleout = 10



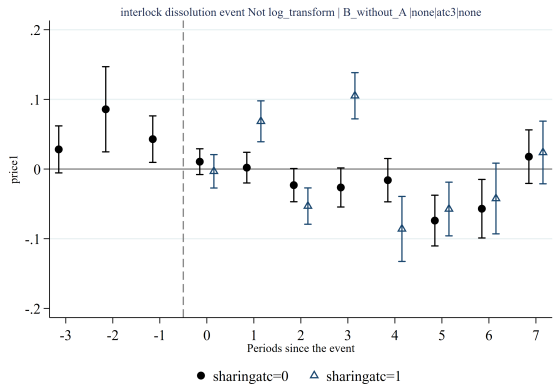
Estimation observations: 88,332

Notes. Black circles report estimates for `sharingatc=0`; hollow blue triangles report estimates for `sharingatc=1`. Error bars are 95% confidence intervals. Within each row, panels differ only in the price1 ruleout setting and resulting sample.

Dissolution after a prior interlock

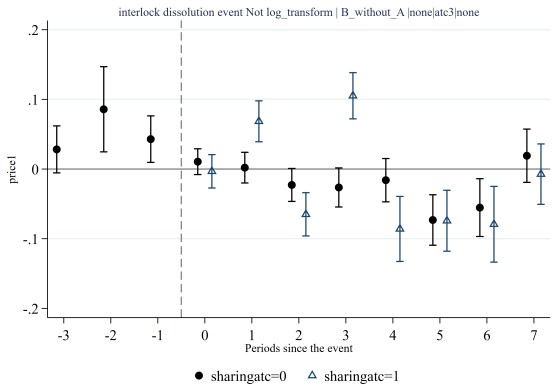
Ruleout = 0

Dynamic sharing



Estimation observations: 66,744

Non-dynamic sharing

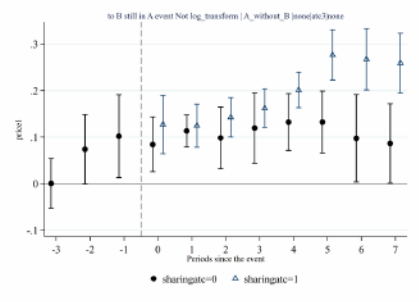


Estimation observations: 66,744

4 To B, Still in A — A-side Treatment

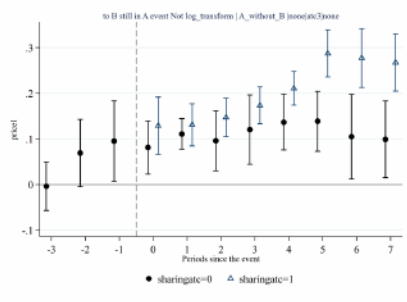
Dynamic sharing

Ruleout = 0



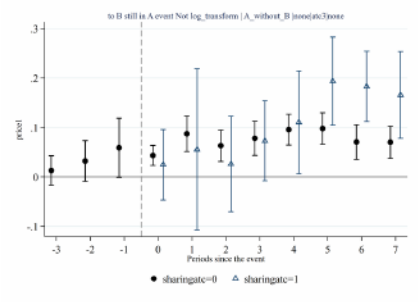
Estimation observations: 108,396

Ruleout = 100



Estimation observations: 107,916

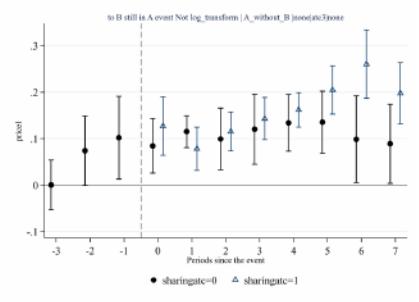
Ruleout = 10



Estimation observations: 104,916

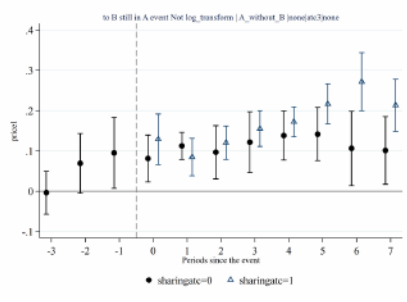
Non-dynamic sharing

Ruleout = 0



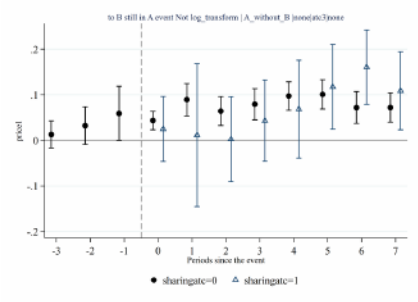
Estimation observations: 108,396

Ruleout = 100



Estimation observations: 107,916

Ruleout = 10



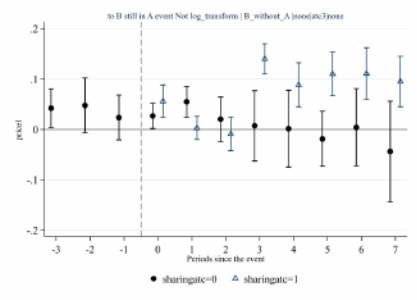
Estimation observations: 104,916

Notes. Black circles report estimates for `sharingatc=0`; hollow blue triangles report estimates for `sharingatc=1`. Error bars are 95% confidence intervals. Within each row, panels differ only in the price1 ruleout setting and resulting sample.

5 To B, Still in A — B-side Treatment

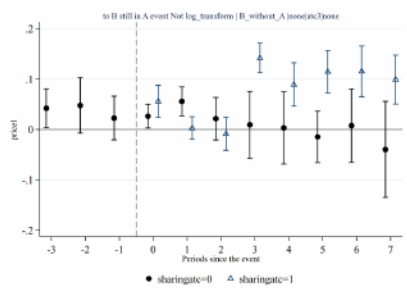
Dynamic sharing

Ruleout = 0



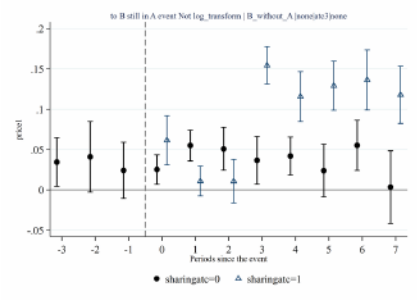
Estimation observations: 130,236

Ruleout = 100



Estimation observations: 129,480

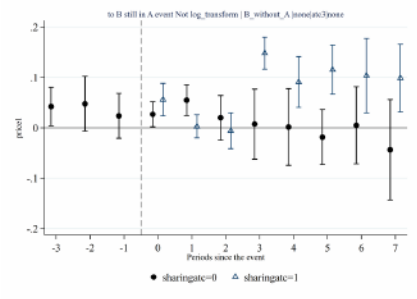
Ruleout = 10



Estimation observations: 119,436

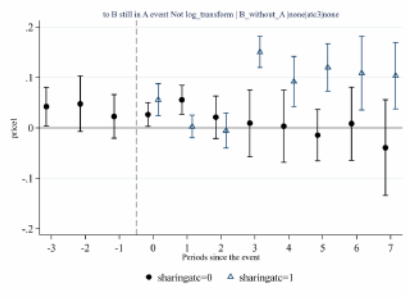
Non-dynamic sharing

Ruleout = 0



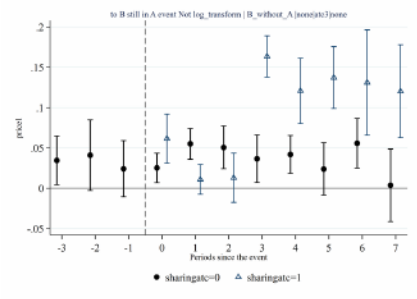
Estimation observations: 130,236

Ruleout = 100



Estimation observations: 129,480

Ruleout = 10



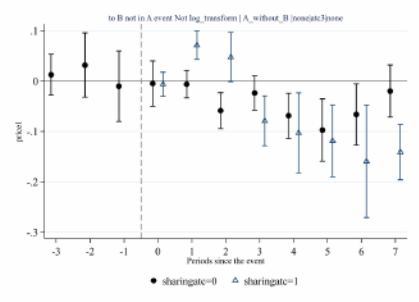
Estimation observations: 119,436

Notes. Black circles report estimates for `sharingatc=0`; hollow blue triangles report estimates for `sharingatc=1`. Error bars are 95% confidence intervals. Within each row, panels differ only in the price1 ruleout setting and resulting sample.

6 To B, Not in A — A-side Treatment

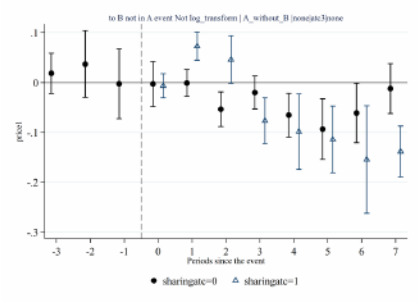
Dynamic sharing

Ruleout = 0



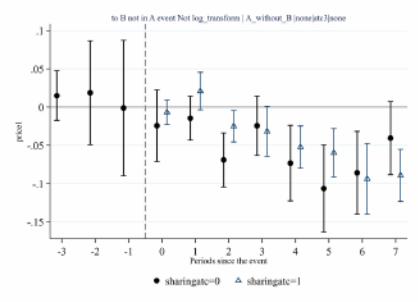
Estimation observations: 73,992

Ruleout = 100



Estimation observations: 73,536

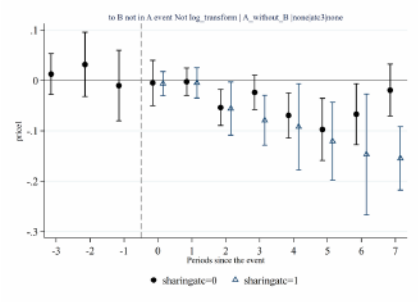
Ruleout = 10



Estimation observations: 70,416

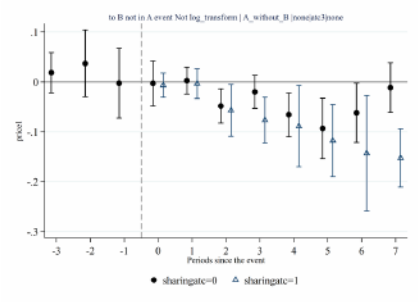
Non-dynamic sharing

Ruleout = 0



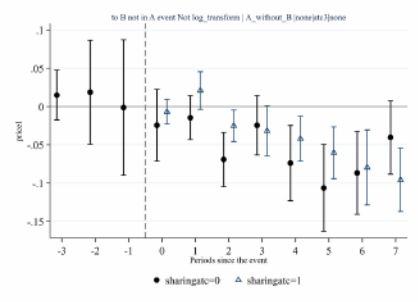
Estimation observations: 73,992

Ruleout = 100



Estimation observations: 73,536

Ruleout = 10



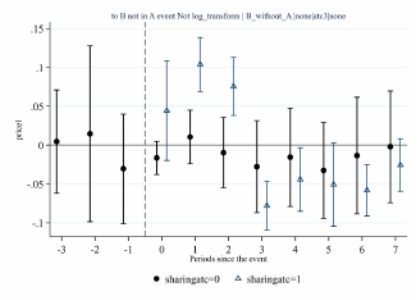
Estimation observations: 70,416

Notes. Black circles report estimates for `sharingatc=0`; hollow blue triangles report estimates for `sharingatc=1`. Error bars are 95% confidence intervals. Within each row, panels differ only in the price1 ruleout setting and resulting sample.

7 To B, Not in A — B-side Treatment

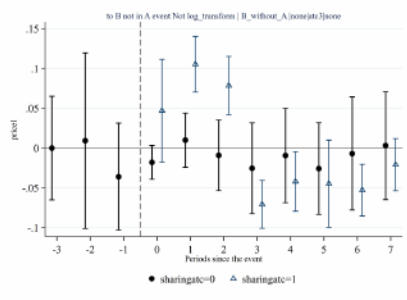
Dynamic sharing

Ruleout = 0



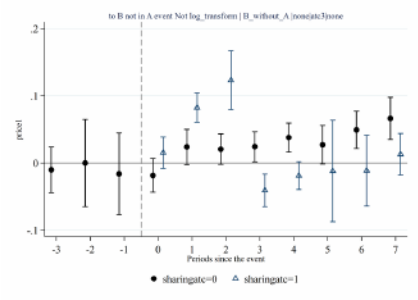
Estimation observations: 66,192

Ruleout = 100



Estimation observations: 65,952

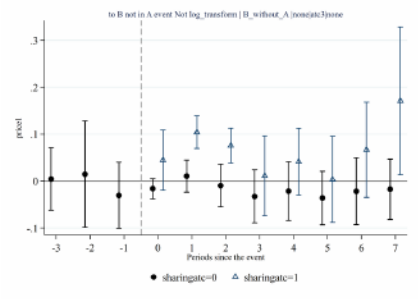
Ruleout = 10



Estimation observations: 62,508

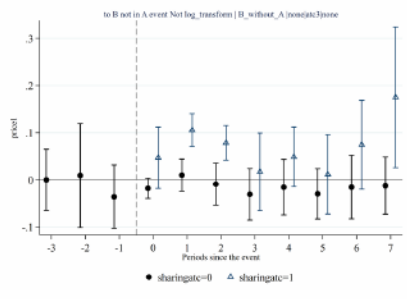
Non-dynamic sharing

Ruleout = 0



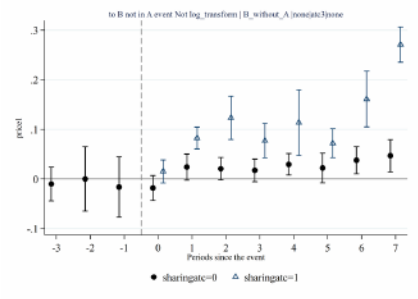
Estimation observations: 66,192

Ruleout = 100



Estimation observations: 65,952

Ruleout = 10



Estimation observations: 62,508

Notes. Black circles report estimates for `sharingatc=0`; hollow blue triangles report estimates for `sharingatc=1`. Error bars are 95% confidence intervals. Within each row, panels differ only in the price1 ruleout setting and resulting sample.